AI Governance, Law-and-Economics

The Ratio Method report: Precedents

The eight frontier AI risks of the *International AI Safety Report 2026*, read against sixteen existing liability regimes

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Introduction

When a new technology causes a new kind of harm, the law rarely starts from nothing. It reaches for a regime it has used before: fault-based tort for car accidents, no-fault funds for vaccine injuries, licensing for nuclear reactors. The hard question is which one.

The Ratio Method, set out in a 2021 law review article,[1] answers that question by breaking liability into a small set of law-and-economics variables. Every category of harm has a profile on those variables. So does every liability regime, because each regime was built for harms of a particular shape. Match the profiles and the precedent follows.

This report applies the method to the eight risks named in the *International AI Safety Report 2026*. In the Ratio Method a risk is always tied to an otherwise ordinary economic activity: AI content crime is a by-product of generative media tools; cyberattacks, of AI-assisted software engineering. Social value is measured for that activity, not for the harm.

The report proceeds in three parts.

- In Variables, we define the eight variables and draw each one. Frequency is split in two, because how often a harm happens and how many people it hits at once call for different legal responses.
- Precedents profiles sixteen historical regimes, from *Rylands v. Fletcher* to the SAFETY Act, on those variables. Together they form a map of how the law has answered each kind of profile before.
- Matches reads each AI risk against that map, variable by variable. It reports which regime each risk most resembles, where the two part ways, and whether the result holds when the matching is made stricter.

Variables

Eight variables describe any category of harm. Six are scales that run from near zero to extreme. Social Value 2 is a set of four categories, and Type of Harm asks three separate questions.

I. Frequency

The 2021 article treats frequency as one variable. This report splits it. Car accidents are frequent, but each one hits a person or two. A reactor accident is vanishingly rare, but hits a region at once. Spread-out losses can be pooled and insured; all-at-once losses cannot, which is why the two have historically produced different regimes.

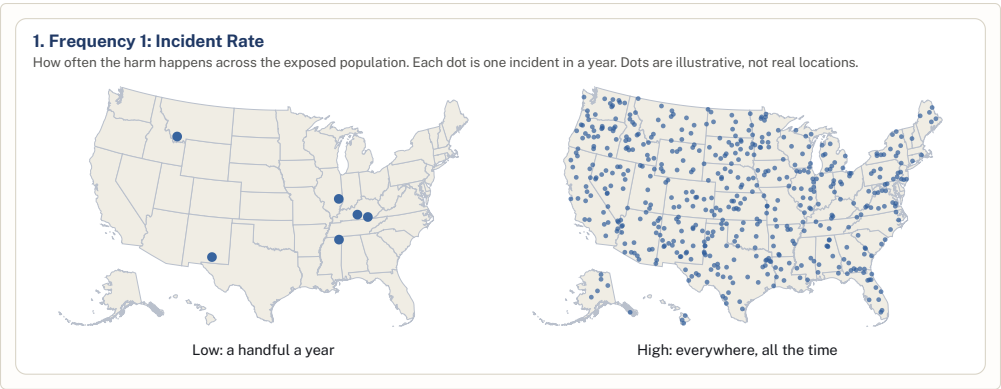


Figure 1.1: Frequency 1 measures how often the harm happens
Each dot is one incident in a year across the exposed population. The dots are illustrative and do not mark real locations.

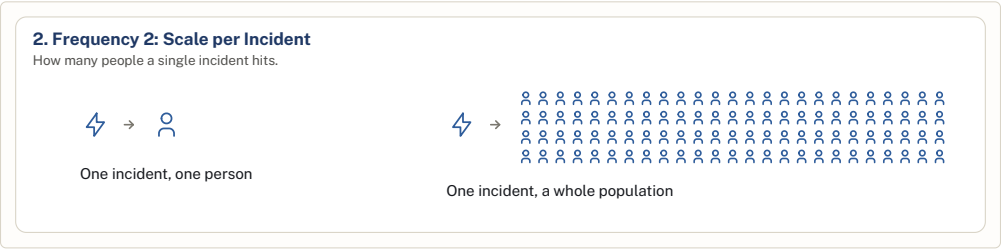


Figure 1.2: Frequency 2 counts the people one incident hits
The same number of incidents can mean one person harmed at a time, or a whole population at once.

II. Average Damages

Average Damages measures how much each victim loses. It is expressed in terms of GDP per capita, so that an incident from 1916, from 2026, or events in different economies, can be read on the same scale.

Examples: A spam call costs a tiny fraction of a year's output. A workplace injury costs about half a year. A death costs more than a hundred.



Figure 1.3: Average Damages measures the harm to each victim
Damages are expressed against GDP per capita, so that harms from different eras and economies can be compared.

III. Social Value

Liability is a tax on an activity, so the method asks what the activity is worth. Social Value 1 is economic: what the activity adds. Social Value 2 is cultural: how society feels about the activity itself, which has historically decided whether a harmful industry is shielded or punished.

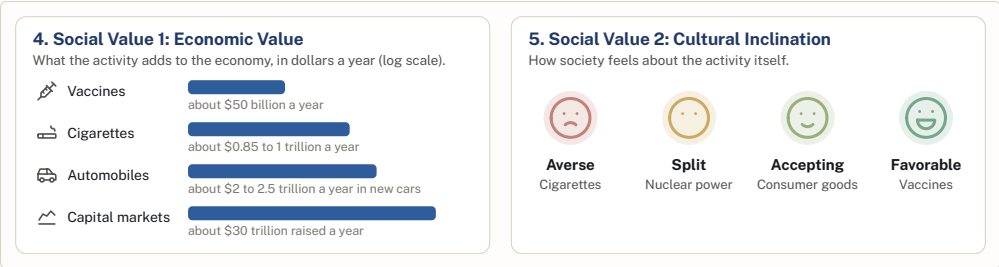


Figure 1.4: Social value has an economic form and a cultural form
Economic value is stated in dollars a year: what the activity sells or raises, not how much good it does. How society regards the activity is the separate question on the right.

IV. Type of Harm

Type of Harm records the mechanism: what kind of injury, how it reaches the victim, and when it shows up. The last mechanism in the figure, a product that acts on its own, is the one the historical regimes do not contain.



Figure 1.5: Type of Harm asks three questions
Injury, mechanism and onset are coded separately. Two harms match on this variable when at least two of the three agree. Spreading and a product acting on its own appear only in the AI risks: no historical regime has either mechanism.

V. Litigation Costs

The last two variables ask whether the courts can handle the harm at all. Litigation Costs 1 weighs the cost of finding out what happened against what the claim is worth. Litigation Costs 2 measures how hard liability is to prove once a claim is brought.

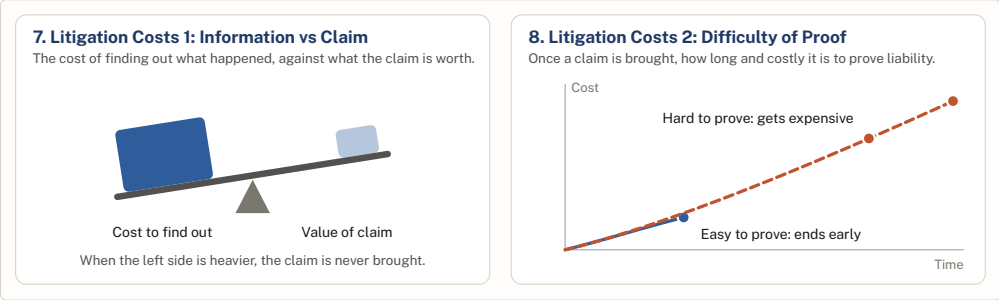


Figure 1.6: Two ways a valid claim can fail
Left: when information costs outweigh the claim, it is never brought. Right: when proof is hard, costs keep climbing after it is.

AI risks of the *International AI Safety Report 2026*

The Report groups its eight risks into three categories (Chapter 2, pp. 44–95). The section and opening page in the Report follow each risk's name.

Malicious use

AI content crime (§2.1.1, p. 45)

Underlying activity: Generative media tools (image, voice, video, text)

Risk: People use these tools to commit scams, fraud and blackmail, and to create non-consensual intimate imagery.

Influence and manipulation (§2.1.2, p. 50)

Underlying activity: AI-generated persuasive content (advertising, marketing, political messaging)

Risk: AI-generated content is used to shift people's beliefs at scale; in experiments it persuades about as well as human writing.

Cyberattacks (§2.1.3, p. 57)

Underlying activity: AI-assisted software engineering and security testing

Risk: Criminal and state-linked groups use the same capabilities to find vulnerabilities and write malicious code.

Biological and chemical risks (§2.1.4, p. 64)

Underlying activity: AI-assisted life-science and chemistry research: designing proteins and drug molecules, and editing genes with tools such as CRISPR

How AI enters: DNA and proteins are sequences, and a sequence can be read and written like code. Models trained on them can propose gene edits and new designs on a computer before anything reaches a lab. In chemistry, models built to screen drug candidates for safety can be pointed the other way, toward toxicity.

Risk: The same models, along with general-purpose assistants that hold expert-level laboratory knowledge, can give novices help toward biological or chemical weapons, such as engineered pathogens or nerve agents.

Malfunctions

Reliability failures (§2.2.1, p. 71)

Underlying activity: AI assistants and agents for professional and everyday tasks

Risk: Systems fabricate facts, write flawed code and give misleading advice, often with no one positioned to catch the error.

Loss of control (§2.2.2, p. 76)

Underlying activity: Development and deployment of highly autonomous frontier AI

Risk: A system comes to operate outside anyone's control with no clear way back; the Report finds current systems cannot yet do this (§2.2.2, p. 76).

Systemic risks

Labour market (§2.3.1, p. 84)

Underlying activity: Automation of cognitive work by employers

Risk: Workers lose jobs and wages to automation, with early signs among early-career workers in exposed occupations.

Human autonomy (§2.3.2, p. 89)

Underlying activity: Consumer AI assistants and companion apps

Risk: Routine reliance weakens critical thinking and breeds uncritical trust in what the system says.

Precedents

Sixteen regimes, in five groups, ordered from the least to the most departure from ordinary tort. Each is tied to the economic activity it governs. The profiles are retrospective readings: legislatures and courts were not reasoning in these variables, and politics shaped every regime listed.

Regime	Underlying activity	What the regime does
Group 1. Ordinary tort.		
Abnormally dangerous activity *(Rylands, 1868)*	Hazardous land use: reservoirs, blasting, storing explosives.	Strict liability on the actor
Strict products liability *(Greenman, 1963; §402A, 1965)*	Mass manufacture and sale of consumer goods.	Strict liability on the manufacturer
Market-share liability *(Sindell, 1980)*	Prescription drug manufacturing: DES, sold by roughly 200 firms.	Liability by market share
Group 2. Aggregation of claims.		
Fraud-on-the-market class action *(Basic, 1988)*	Issuing and trading public securities.	Presumption + class aggregation
Statutory damages/public enforcer *(FTC Act, 1914; TCPA, 1991)*	Telemarketing and mass consumer marketing.	Fixed damages; agency sues
Tobacco Master Settlement *(1998)*	Manufacture and sale of cigarettes.	States recover public costs
CERCLA / Superfund *(1980)*	Industrial production and chemical waste disposal.	Retroactive strict liability + fund
Group 3. No-fault funds.		
Workers' compensation *(Germany 1884; US 1910s)*	Industrial employment: factories, railroads, mines.	No-fault benefits, employer-funded
Vaccine injury compensation *(1986)*	Childhood vaccine manufacture and mass immunisation.	No-fault fund; maker shielded
Displacement adjustment *(TAA, 1962; WARN, 1988)*	Trade liberalisation and plant closings.	Notice, retraining, income support
Group 4. Catastrophic risk.		
Ex ante licensing *(FDA, NRC, FAA, Select Agent rules)*	Marketing new drugs; operating reactors; certifying aircraft and airlines; handling dangerous pathogens.	Permission before the act
Price-Anderson (Nuclear Liability) *(1957)*	Commercial nuclear power generation.	Channeling, insurance, pool, cap
Oil spill liability *(Oil Pollution Act, 1990)*	Transporting and drilling for oil at sea.	Strict liability capped by carrier size; uncapped for gross negligence; proof of financial responsibility; trust fund
Space Liability Convention *(1972)*	Government and commercial space launch.	Absolute liability on launching state
Group 5. Shields.		
Intermediary immunity *(CDA s.230, 1996)*	Hosting user-generated content online.	Intermediary immune; third party liable
SAFETY Act *(2002)*	Selling anti-terrorism and security technology.	Liability capped at insurance level

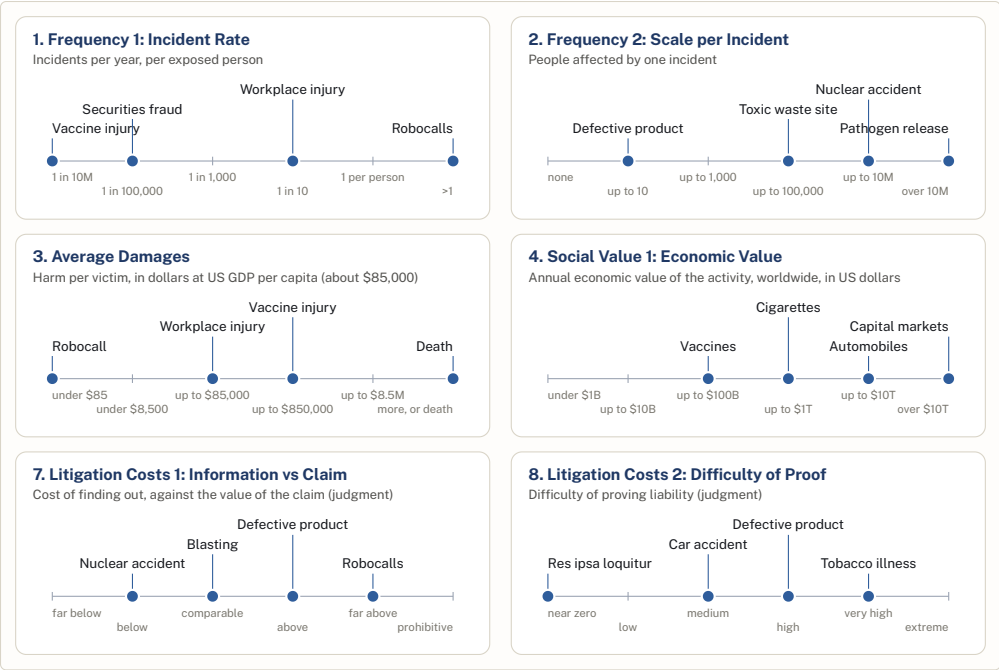


Figure 2.1: Familiar harms on each scale

The six scaled variables, each marked in its own unit, with familiar harms and doctrines placed on them. Dollar figures use a US GDP per capita of about $85,000. Scales marked judgment have no natural unit and are coded in six ordered steps.

The frequency map

Plotting the two forms of frequency against each other shows why the split matters. Car-accident-style harms sit low and to the right. Catastrophic harms sit high and to the left. Loss of control and biological and chemical risks sit at the top of the first column, directly above the regimes built for catastrophe.

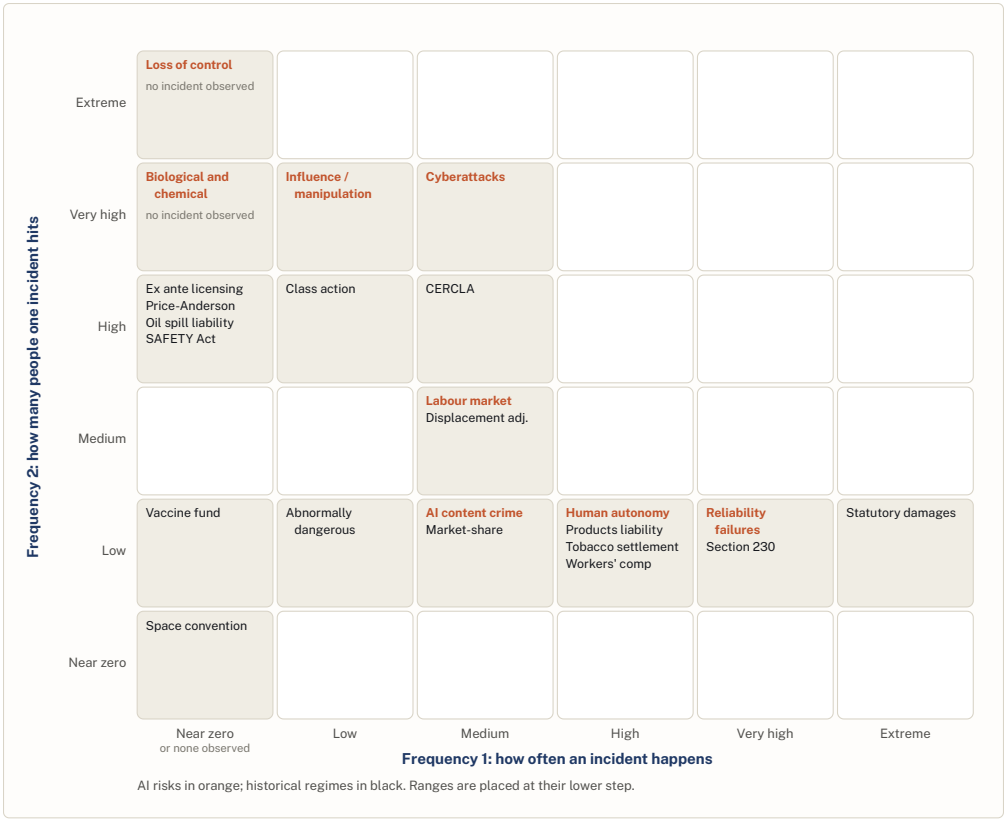


Figure 2.2: AI risks sit beside the historical harms they most resemble
Frequency 1 against Frequency 2. AI risks in orange, historical regimes in black. Ranges are placed at their lower step. Biological and chemical risks and loss of control sit in the first column because no incident has been observed, not because one is rare; their severity shows on the vertical axis.

Matches

Each risk is compared, variable by variable, against every regime. A variable matches when the two sit on the same step of its scale or one step apart. Regimes are ranked by matches minus misses, with equal weights, and every result is re-run under strict matching, where only the same step counts. The full rules are in the Appendix.

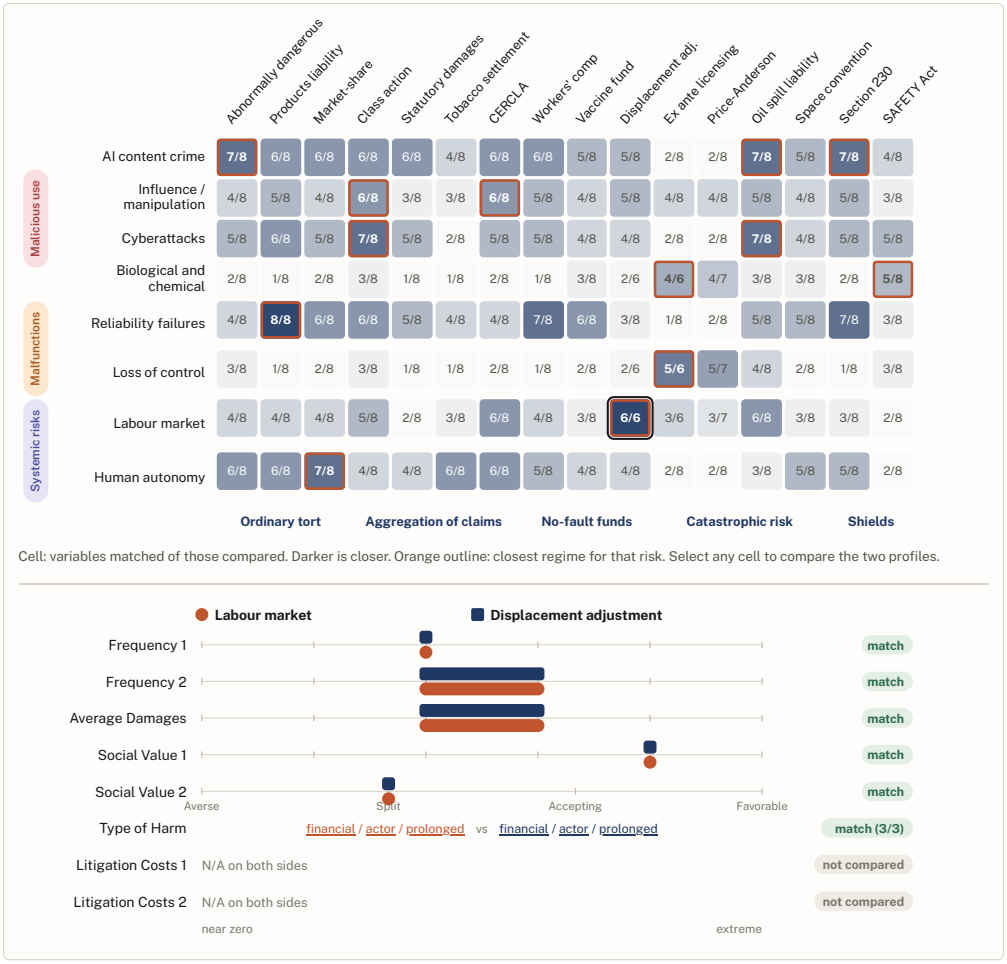


Figure 3.1: The catastrophic risks match the regimes built for catastrophe and almost nothing else

Every AI risk against every historical regime. Select a cell to see the risk (orange) against the regime (navy), variable by variable. A grey link joins the two where they match. For Type of Harm, the parts the two share are underlined.

Risk by risk

AI content crime

Underlying activity	Generative media tools: image, voice, video and text generation.
Closest regime	Three-way tie: abnormally dangerous activity; oil spill liability; intermediary immunity (s.230) *(7 of 8 each)*
Where it diverges	Abnormally dangerous activity: Type of Harm (a third party does the harm). Oil spill liability: Frequency 2. Section 230: Social Value 2.
Holds under strict matching?	No. Strict products liability leads at 6 of 8; Section 230 is second.
Output	**No stable analogue. Candidates: strict liability on the tool's provider; intermediary immunity; strict products liability.**

Influence / manipulation

Underlying activity	AI-generated persuasive content: advertising, marketing, political messaging.
Closest regime	Tie: fraud-on-the-market class action; CERCLA *(6 of 8 each)*
Where it diverges	Fraud-on-the-market: Social Value 2 and Type of Harm (no priced loss to measure). CERCLA: Average Damages and Type of Harm.
Holds under strict matching?	Partly. Fraud-on-the-market leads alone at 5 of 8.
Output	**Presumption of causation + aggregated claim, brought by a public enforcer.**

Cyberattacks

Underlying activity	AI-assisted software engineering and security testing.
Closest regime	Tie: fraud-on-the-market class action; oil spill liability *(7 of 8 each)*
Where it diverges	Class action: Litigation Costs 2 (a breach is easier to prove than reliance). Oil spill liability: Social Value 2.
Holds under strict matching?	Partly. The class action leads alone at 6 of 8; oil spill liability falls to 4 of 8.
Output	**Aggregated claims with a presumption of harm; or oil-spill-style liability: strict, capped by the carrier's size, with proof of financial responsibility and a trust fund. The second is what the 2021 article proposed.**

Biological and chemical

Underlying activity	AI-assisted biological and chemical research.
Closest regime	Tie: ex ante licensing; SAFETY Act *(4 of 6; 5 of 8)*
Where it diverges	Type of Harm on both: the harm spreads on its own, and no historical regime has that mechanism. Licensing also misses Social Value 2; the SAFETY Act misses both Litigation Costs columns, which it loses only because the risk is coded N/A there, see Limitations.
Holds under strict matching?	Yes. Licensing leads alone under strict matching.
Output	**Permission before release, with a certified liability cap for developers whose safeguards are approved.**

Reliability failures

Underlying activity	AI assistants and agents used for professional and everyday tasks.
Closest regime	Strict products liability *(8 of 8)*
Where it diverges	None under adjacent matching. One step apart on Frequency 1, Social Value 1 and Litigation Costs 1.
Holds under strict matching?	No. Section 230 leads at 6 of 8; products liability is second at 5 of 8.
Output	**Strict liability on the developer for defects. Tort generally does not compensate pure economic loss, so contract and warranty still govern most claims.**

Loss of control

Underlying activity	Developing and deploying highly autonomous frontier AI systems.
Closest regime	Ex ante licensing *(5 of 6)*
Where it diverges	Type of Harm: the product itself is the agent, which no historical regime contemplates. Litigation Costs are not compared (N/A on both sides); the N/A coding favours licensing on these two columns, see Limitations.
Holds under strict matching?	Yes.
Output	**Permission before deployment. Price-Anderson is second (5 of 7) and suggests a pool for the residual.**

Labour market

Underlying activity	Employers automating cognitive work.
Closest regime	Displacement adjustment *(6 of 6)*
Where it diverges	None on the compared columns. Litigation Costs not compared (N/A on both sides). Differs on who pays.
Holds under strict matching?	Yes.
Output	**Notice, retraining, income support, with workers' comp financing (employer-funded), per Section IV.C of the article.**

Human autonomy

Underlying activity	Consumer AI assistants and companion apps.
Closest regime	Market-share liability *(7 of 8)*
Where it diverges	Social Value 2: prescription drugs are accepted; consumer AI companions divide opinion.
Holds under strict matching?	No. The tobacco settlement and workers' compensation tie at 5 of 8; market-share liability drops to 4 of 8.
Output	**Liability shared across providers by market share. Not stable: tobacco-style recovery of public costs is the other candidate.**

Discussion

Five of the eight results hold under strict matching: manipulation, cyberattacks, biological and chemical risks, loss of control and labour market keep a leading regime when only same-step matches count. Three do not. AI content crime, reliability failures and human autonomy change their leading regime, and their outputs should be read as candidates.

How much the method discriminates. Read across the scorecard, the method does two jobs with different reliability. It separates the catastrophic risks from everything else robustly: biological and chemical risks and loss of control match licensing and the catastrophe regimes, and nothing in ordinary tort, under every matching rule and weighting tried. Within the rest it discriminates weakly. Eight equal-weight columns with one-step adjacency leave a three-way tie for content crime and two-way ties for manipulation and cyberattacks, and three leaders change under strict matching. A weight-sensitivity check makes this concrete. Re-ranking under six alternative weightings (Type of Harm doubled; Average Damages doubled; both doubled; both Frequency columns doubled; Litigation Costs halved; Social Value 2 dropped) never changes the leader for the four risks that have a single one: reliability failures, loss of control, labour market and human autonomy, though dropping Social Value 2 brings Section 230 level with products liability for reliability failures. For the four tied risks the weightings break the ties in different directions, almost always in favour of a regime already in the tie; the exceptions are manipulation, where doubling Type of Harm adds Section 230 to the tie and halving Litigation Costs adds displacement adjustment and oil spill liability. Weights decide between near neighbours; they do not produce new matches. The outputs for the tied risks should be read as a short list, not a selection.

Tying each risk to its underlying activity moved three results. Cyberattacks tie between the class action, which is how data-breach claims are in fact litigated, and oil spill liability, which is the regime the 2021 article proposed for them: data, like oil, is a valuable resource that does region-wide harm under careless oversight. Reliability failures match strict products liability on all eight variables. Biological and chemical risks tie licensing with the SAFETY Act, which suggests permission before release paired with a certified liability cap for developers whose safeguards are approved.

Labour market matches displacement adjustment on all six compared variables. It is also the risk the 2021 article addressed directly, in its proposal for employer-funded retraining.

Type of Harm and Social Value 2 are the most frequent misses. Third-party wrongdoers do appear in the historical regimes, in Section 230 and the SAFETY Act. What has no precedent are the two mechanisms coded for the catastrophic risks: a harm that spreads on its own, as an engineered pathogen would, and a

product that acts as the agent, as in loss of control. Both risks miss on Type of Harm against every regime in the map.

Limitations

- **Figures.** The numbers come from different years, countries and populations. Several are not specific to AI: the FBI and ITRC counts cover all internet crime and all data breaches, and AI's share is not reported. They fix the order of magnitude, not the ratio.
- **Sources.** The Appendix lists the source for each figure. Every figure was cite-checked against its source in October 2026; nine were tightened as a result and are marked in the table (the smoker count and NPL count updated to current data; the securities-filing range widened; the TAA start year, the sulfanilamide and thalidomide attributions, the Three Mile Island evacuation source and the 9/11 death-toll source corrected; the employment-decline figure dated). Citations to the Report give section and opening page; the eight risks are defined at §§ 2.1.1–2.3.2 (pp. 45–95).
- **Weights.** Every column counts equally. The checks drop one column at a time and re-rank under six alternative weightings (see "How much the method discriminates" above). The leader is stable where there is one and the ties are weight-dependent. Alternative classifications of the regimes have not been tested.
- **The N/A rule.** Litigation Costs are coded N/A for biological and chemical risks and for loss of control, on the ground that liability after the fact is not a meaningful remedy. That coding is not neutral. Under the matching rules a shared N/A drops the column and a one-sided N/A counts as a miss, so every ex post regime loses two columns against these two risks before any comparison is made, while licensing and displacement adjustment, the two regimes also coded N/A, lose none. The N/A coding therefore pre-commits part of the result toward ex ante licensing for exactly the two risks where licensing wins. The alternative is to code Litigation Costs for these risks on their merits (very high: causation across a spreading harm; extreme: no plaintiff may remain to sue) and compare on all eight columns. Whether licensing still leads under that coding has not been tested; the Type of Harm result, on which both risks miss against every regime, does not depend on it.
- **Average Damages basis.** Damages are per victim. For data breaches that means per notified person, which is far smaller than the cost to the breached organisation.
- **Type of Harm mapping.** Both tables need Type of Harm recast onto the six axes. Two gaps already show: lawful displacement has no reason-for-injury value, and psychological harm has no type-of-injury value.
- **Common cause.** Reliability failures are spread out but trace to one model, as DES traced to one product. This belongs in Type of Harm as a categorical axis, not in Frequency.

Appendix

▼ Matching rules

Rule	How it is applied
Underlying activity	Each regime and each risk is tied to the ordinary economic activity that produces the harm. Social Value 1 and 2 describe that activity, not the harm.

Rule	How it is applied
Frequency 1 steps	Incidents per year, relative to the exposed population. 0: none observed, or under 1 in 10 million. 1: up to 1 in 100,000. 2: up to 1 in 1,000. 3: up to 1 in 10. 4: up to 1 per person. 5: more than 1 per person.
Frequency 2 steps	People affected per incident. 0: none. 1: up to 10. 2: up to 1,000. 3: up to 100,000. 4: up to 10 million. 5: more than 10 million.
Average Damages steps	Harm per victim, relative to GDP per capita. 0: under 0.1%. 1: under 10%. 2: up to 1 times. 3: up to 10 times. 4: up to 100 times. 5: more, including death.
Other columns	Social Value 1 and both Litigation Costs columns are coded by judgment on the same six steps: near zero, low, medium, high, very high, extreme. Where no count exists, the cell says "judgment".
Match	A column matches when the risk and the regime sit on the same step or one step apart. A range such as "low to medium" counts as both steps.
Social Value 2	Coded in four categories: averse, split, accepting, favorable. Matches only on the same category.
Type of Harm	Coded on three parts: kind of injury, who or what does the harm, and onset. Matches when at least two of the three agree. Two mechanisms, spreading (a harm that propagates on its own) and a product acting on its own, appear in AI risks only; no historical regime is coded with either.
"N/A"	N/A on both sides: not compared, because a shared N/A reflects one legal fact, not independent similarity. N/A on one side only: a miss. This rule is not neutral between regimes; see "The N/A rule" under Limitations.
Ranking	Regimes are ranked by matches minus misses. Every compared column carries equal weight. That is a default, not a finding (see Open Items).
Robustness	Each result is re-run under strict matching (same step only) and with each column dropped in turn.

▼ Historical regimes: activity, frequency and damages

Regime	Underlying activity	Frequency 1: incident rate	Frequency 2: scale per incident	Average damages
Group 1. Ordinary tort.				
Abnormally dangerous activity *(Rylands, 1868)*	Hazardous land use: reservoirs, blasting, storing explosives.	Low (judgment). No national count; claims arise case by case.	Low to medium: one neighbouring mine owner in Rylands; a neighbourhood in a blast.	High (judgment): a flooded mine, a destroyed building.
Strict products liability *(Greenman, 1963; §402A, 1965)*	Mass manufacture and sale of consumer goods.	High: 20 million people injured a year in a population of about 205 million, roughly 1 in 10 (National Commission on Product Safety, 1970).	Low: one user per incident; the defect is shared across the product line.	Medium for the typical injury. Of the 20 million, 110,000 were permanently disabled and 30,000 killed.
Market-share liability *(Sindell, 1980)*	Prescription drug manufacturing: DES, sold by roughly 200 firms.	Medium: about 1 in 1,000 exposed daughters developed clear-cell cancer; 5 to 10 million people were exposed, 1938 to 1971.	Low: one patient per incident; the product is shared.	High to very high: cancer diagnosed in young adulthood.
Group 2. Aggregation of claims.				
Fraud-on-the-market class action *(Basic, 1988)*	Issuing and trading public securities.	Low: roughly 200 to 230 securities class actions are filed a year.	High to very high: thousands to millions of shareholders per misstatement.	Low: a small fraction of each investor's holding.
Statutory damages/public enforcer *(FTC Act, 1914; TCPA, 1991)*	Telemarketing and mass consumer marketing.	Extreme: 52.8 billion robocalls in the US in 2024, nearly 200 per adult (YouMail).	Low: one recipient per call.	Tiny: a nuisance call. The statute fixes $500 per violation, $1,500 if wilful.

Regime	Underlying activity	Frequency 1: incident rate	Frequency 2: scale per incident	Average damages
Tobacco Master Settlement *(1998)*	Manufacture and sale of cigarettes.	High: about 480,000 US deaths a year; 28.8 million adults smoked in 2022, down from roughly 45 million in the mid-2000s, so on the order of 1 to 2 in 100 (CDC).	Low: one smoker at a time; the product is shared.	Very high to extreme: chronic disease and death. The settlement: $206 billion over 25 years to 46 states.
CERCLA / Superfund *(1980)*	Industrial production and chemical waste disposal.	Medium (judgment): about 1,340 sites on the National Priorities List.	High: a community per site; about 950 families were relocated from Love Canal.	High (judgment): homes lost, long-term health effects.
Group 3. No-fault funds.				
Workers' compensation *(Germany 1884; US 1910s)*	Industrial employment: factories, railroads, mines.	High: about 23,000 industrial deaths in 1913 among 38 million US workers (61 per 100,000); non-fatal injuries were far more common.	Low: one worker per incident.	Medium: lost wages and medical costs for the typical injury.
Vaccine injury compensation *(1986)*	Childhood vaccine manufacture and mass immunisation.	Near zero: about 1 person compensated per 1 million doses distributed (HRSA).	Low: one patient per incident.	High: serious, sometimes permanent injury.
Displacement adjustment *(TAA, 1962; WARN, 1988)*	Trade liberalisation and plant closings.	Medium: more than 5 million workers served by Trade Adjustment Assistance since 1974, on the order of 100,000 a year.	Medium to high: WARN covers layoffs of 50 or more at employers of 100 or more; large closings reach thousands.	Medium to high: lasting earnings losses.
Group 4. Catastrophic risk.				
Ex ante licensing *(FDA, NRC, FAA, Select Agent rules)*	Marketing new drugs; operating reactors; certifying aircraft and airlines; handling dangerous pathogens.	Near zero: rare events.	High to extreme: thousands of children affected by thalidomide worldwide, about 10,000 by the standard estimate; a reactor or pathogen release could reach far more. (The 1938 Act followed more than 100 deaths from Elixir Sulfanilamide.)	Extreme: death and birth defects.
Price-Anderson (Nuclear Liability) *(1957)*	Commercial nuclear power generation.	Near zero: one major US accident (Three Mile Island, 1979).	High to very high: the 1957 federal study (WASH-740) projected up to 3,400 deaths and 43,000 injuries; about 140,000 people left the area around Three Mile Island.	Extreme as projected: deaths, and $7 billion in property damage in 1957 dollars. Actual Three Mile Island claims: about $71 million in total, no deaths.
Oil spill liability *(Oil Pollution Act, 1990)*	Transporting and drilling for oil at sea.	Near zero to low (judgment): major spills are rare. The Act followed the Exxon Valdez spill of 1989.	High to very high: more than 32,000 claimants in the Exxon Valdez punitive-damages class; Deepwater Horizon reached the coastal economies of five states.	Low to medium: 0.6% of GDP per capita when spread across the affected states (the 2021 article); far more for fishing and tourism businesses.
Space Liability Convention *(1972)*	Government and commercial space launch.	Near zero: one claim since 1972 (Cosmos 954, 1978).	None to low: no one was injured; debris fell across northern Canada.	High, borne by a state: Canada claimed C$6 million and settled for C$3 million.

Regime	Underlying activity	Frequency 1: incident rate	Frequency 2: scale per incident	Average damages
Group 5. Shields.				
Intermediary immunity *(CDA s.230, 1996)*	Hosting user-generated content online.	Very high (judgment): Prodigy alone carried about 60,000 postings a day for 2 million subscribers in 1995.	Low: one person defamed or defrauded per posting.	Low to medium (judgment).
SAFETY Act *(2002)*	Selling anti-terrorism and security technology.	Near zero: one attack on the scale of 11 September 2001.	High: 2,977 killed and thousands injured.	Extreme: death.

▼ Historical regimes: social value, type of harm, litigation costs and output

Regime	Social value 1	Social value 2	Type of harm	Litigation costs 1	Litigation costs 2	Output
Group 1. Ordinary tort.						
Abnormally dangerous activity *(Rylands, 1868)*	Moderate	Split: tolerated, uncommon	Physical/property, proximate	Moderate	High: care was taken, no fault to prove	Strict liability on the actor
Strict products liability *(Greenman, 1963; §402A, 1965)*	High	Accepting of mass-produced goods	Physical, via product	High: evidence inside the factory	High: negligence unprovable	Strict liability on the manufacturer
Market-share liability *(Sindell, 1980)*	High: prescription drugs	Accepting: medicine is valued	Physical, onset delayed a generation	High	Very high: can't identify which maker	Liability by market share
Group 2. Aggregation of claims.						
Fraud-on-the-market class action *(Basic, 1988)*	High: capital markets	Accepting: public investing is ordinary	Financial, intangible	High: claim cost exceeds individual loss	Very high: individual reliance	Presumption + class aggregation
Statutory damages/public enforcer *(FTC Act, 1914; TCPA, 1991)*	Medium	Averse: unsolicited calls are widely resented	Financial or rights, intangible	Very high: claim cost exceeds recovery	Moderate: violation easy to show	Fixed damages; agency sues
Tobacco Master Settlement *(1998)*	Low	Averse: by 1998 public opinion had turned against smoking	Physical, prolonged	High	Very high: individual causation, assumed risk	States recover public costs
CERCLA / Superfund *(1980)*	High: industrial production	Split: industry valued, dumping feared	Environmental, prolonged	High: decades-old records	Very high: many or defunct defendants	Retroactive strict liability + fund
Group 3. No-fault funds.						
Workers' compensation *(Germany 1884; US 1910s)*	Very high	Accepting of industrialization	Physical, workplace, immediate	High: worker vs employer	Very high: fault defenses beat most claims	No-fault benefits, employer-funded
Vaccine injury compensation *(1986)*	Very high	Strongly favorable	Physical, via product	High	High: medical causation	No-fault fund; maker shielded
Displacement adjustment *(TAA, 1962; WARN, 1988)*	Very high: gains from trade	Split: trade valued, closings resented	Financial, prolonged	N/A: no claim exists	N/A: no wrongful act	Notice, retraining, income support

Regime	Social value 1	Social value 2	Type of harm	Litigation costs 1	Litigation costs 2	Output
Group 4. Catastrophic risk.						
Ex ante licensing *(FDA, NRC, FAA, Select Agent rules)*	High	Split: valued but feared	Physical, mass, spanning	N/A	N/A	Permission before the act
Price-Anderson (Nuclear Liability) *(1957)*	High	Split; the state wanted the industry	Physical, mass, spanning	Low: cause obvious	N/A: problem is solvency, not proof	Channeling, insurance, pool, cap
Oil spill liability *(Oil Pollution Act, 1990)*	Very high: oil and gas	Split: fuel valued, spills feared	Environmental + financial, immediate	Moderate: the source of a spill is usually evident	Moderate: strict liability; the responsible party is known	Strict liability capped by carrier size; uncapped for gross negligence; proof of financial responsibility; trust fund
Space Liability Convention *(1972)*	High	Favorable	Physical, cross-border	High: foreign actor	High: jurisdiction	Absolute liability on launching state
Group 5. Shields.						
Intermediary immunity *(CDA s.230, 1996)*	Very high	Strongly favorable (speech)	Rights or financial, intangible, third-party agent	High: anonymous third party	High	Intermediary immune; third party liable
SAFETY Act *(2002)*	High: defensive technology	Favorable	Physical, third-party malice	Moderate	Moderate	Liability capped at insurance level

▼ AI risks: activity, frequency and damages

Risk	Underlying activity	Frequency 1: incident rate	Frequency 2: scale per incident	Average damages
AI content crime	Generative media tools: image, voice, video and text generation.	Medium to high. AI-specific counts are partial: 67,000 reports of generative-AI child exploitation to NCMEC in 2024, up from 4,700 in 2023. All US internet crime: 859,532 FBI complaints in 2024; the AI share is not reported.	Low: one victim per incident; a campaign repeats the incident.	Low to medium: the average loss per FBI complaint in 2024 was $19,372, about a quarter of US GDP per capita. Dignitary harm is unpriced.
Influence / manipulation	AI-generated persuasive content: advertising, marketing, political messaging.	Low to medium (judgment). The Report (§2.1.2, p. 50): documented, not yet widespread. No incident count exists.	Very high (judgment): a campaign addresses a population; reach is unmeasured.	Tiny to low per person; unmeasured.
Cyberattacks	AI-assisted software engineering and security testing.	Medium: 3,158 US data compromises in 2024 (ITRC), among several million firms. The AI share is not reported. The Report cites an AI agent finding 77% of the vulnerabilities in one competition (§2.1.3, p. 57).	Very high: 1.7 billion victim notices in 2024, about 540,000 per compromise; six breaches exceeded 100 million each.	Low to medium per notified person. Per breached organisation: $4.88 million on average (IBM, 2024).

Risk	Underlying activity	Frequency 1: incident rate	Frequency 2: scale per incident	Average damages
Biological and chemical	AI-assisted biological and chemical research.	None observed.	Very high to extreme: a chemical release could reach thousands; the reference class for an engineered pathogen is COVID-19, about 7 million reported deaths (WHO).	Extreme: death.
Reliability failures	AI assistants and agents used for professional and everyday tasks.	Very high (judgment): errors are routine across about 800 million weekly users of one assistant alone (OpenAI, 2025). No incident count exists.	Low: one user per error; the cause is shared across the model.	Low to medium. Examples: a US$5,000 court sanction for fabricated citations (Mata v. Avianca, 2023); C$812 awarded against an airline for its chatbot's advice (Moffatt v. Air Canada, 2024).
Loss of control	Developing and deploying highly autonomous frontier AI systems.	None observed. This is a historical count, not a forecast.	Extreme (judgment): by definition, not contained.	Extreme: unbounded.
Labour market	Employers automating cognitive work.	Medium (judgment). The Report (§2.3.1, p. 84): no effect on overall employment so far. One study: a 13% relative decline in employment for workers aged 22 to 25 in the most AI-exposed occupations, as of July 2025 data (Brynjolfsson, Chandar & Chen, 2025); the authors' August 2026 update puts it at 16% to 19%.	Medium to high: a team or cohort per deployment.	Medium to high: lasting earnings losses.
Human autonomy	Consumer AI assistants and companion apps.	High: tens of millions of companion-app users (the Report, §2.3.2, p. 89). OpenAI estimates 0.15% of weekly users show heightened emotional attachment, about 1.2 million people at 800 million weekly users.	Low: one user per incident; the product is shared.	Low to medium; unmeasured.

▼ AI risks: social value, type of harm and litigation costs

Risk	Social value 1	Social value 2	Type of harm	Litigation costs 1	Litigation costs 2
AI content crime	High: used across media, advertising and design	Split: popular; contested over deepfakes and creators' rights	Financial + dignitary, intangible, third-party malice	High: anonymous perpetrator, claim cost exceeds loss	High: developer remote from the harm
Influence / manipulation	High	Split: persuasion is ordinary and protected; AI-made persuasion is distrusted	Rights, intangible, prolonged	Very high	Very high: causation near-unprovable
Cyberattacks	High: software underpins most industries	Accepting of automated coding and security tools	Financial, intangible, third-party malice	High: attribution	Moderate
Biological and chemical	High: drug discovery and biomedical research	Favorable: medical research is strongly valued	Biological + physical harm; spreading; onset uncertain	N/A ex post	N/A ex post

Risk	Social value 1	Social value 2	Type of harm	Litigation costs 1	Litigation costs 2
Reliability failures	Very high	Accepting: tolerant of errors so far	Financial, via product, immediate	Moderate: failure visible	High: fault is inside the model
Loss of control	High	Split: wanted for productivity, feared for autonomy	Physical + financial harm; the product acts on its own; onset uncertain	N/A ex post	N/A ex post
Labour market	Very high: productivity gains	Split: gains valued, job loss feared	Financial, prolonged	N/A: no claim exists	N/A: no wrongful act
Human autonomy	Moderate	Split	Psychological, prolonged	High	Very high: causation and consent

▼ Sources for the figures cited

Figure	Source	Status
20 million injured, 110,000 disabled, 30,000 killed a year	National Commission on Product Safety, Final Report (1970)	Checked, Sept 2026
859,532 complaints; $16.6 billion; $19,372 average loss	FBI Internet Crime Complaint Center, 2024 Annual Report	Checked, Sept 2026
4,700 (2023) and 67,000 (2024) generative-AI reports	National Center for Missing & Exploited Children, CyberTipline data	Checked, Sept 2026
3,158 compromises; 1.7 billion victim notices	Identity Theft Resource Center, 2024 Annual Data Breach Report (as updated)	Checked, Sept 2026
52.8 billion robocalls; nearly 200 per adult	YouMail Robocall Index, 2024	Checked, Sept 2026
77% of vulnerabilities; tens of millions of companion users; no effect on overall employment; manipulation documented, not widespread	The Report, executive summary (p. 11) and §§ 2.1.2 (p. 50), 2.1.3 (p. 57), 2.3.1 (p. 84), 2.3.2 (p. 89)	Checked, Sept 2026; section pinpoints added Oct 2026
Tobacco: about $0.85 to 1 trillion a year worldwide	MarketLine, Tobacco & Tobacco Products Global Industry Guide (2024: $844 billion); Statista (2024: $965 billion)	Checked, Sept 2026
Automobiles: about $2 to 2.5 trillion a year in new car sales worldwide	MarketLine, New Cars Global Industry Guide (2024: $2.44 trillion); Statista (2024: $1.98 trillion)	Checked, Sept 2026
Vaccines: about $50 billion a year worldwide, excluding COVID-19	Industry market estimates for 2024; WHO Global Vaccine Market Report for market structure	Checked, Sept 2026
Capital markets: about $30 trillion raised a year worldwide; $157.8 trillion equity market capitalization	SIFMA, Capital Markets Fact Book 2026 (2025 data)	Checked, Sept 2026
$1.3 trillion of US economic activity from oil and gas (2015); Deepwater Horizon ratios	PwC for the American Petroleum Institute (2017) and the Sample Data Set, both as cited in Margolin & Frazier (2021)	From the article
More than 32,000 Exxon Valdez claimants (the punitive-damages class)	Exxon Shipping Co. v. Baker, 554 U.S. 471, 479 & n.6 (2008)	Checked, Oct 2026
23,000 industrial deaths among 38 million workers (1913)	CDC, "Improvements in Workplace Safety," 48 MMWR 461 (June 11, 1999), citing Bureau of Labor Statistics	Checked, Oct 2026
480,000 deaths a year; 28.8 million adult smokers (2022); about 45 million in the mid-2000s	CDC: 2014 Surgeon General's Report (deaths); "Cigarette Smoking in the United States," 2022 data (smokers)	Checked, Oct 2026 (smoker count updated)
$206 billion over 25 years; 46 states	Master Settlement Agreement (1998); GAO, Tobacco Settlement, GAO-01-851 (2001)	Checked, Oct 2026
5 to 10 million exposed; about 1 in 1,000; roughly 200 firms	NCI, DES Exposure and Cancer fact sheet; Sindell v. Abbott Laboratories, 26 Cal. 3d 588, 602 (1980)	Checked, Oct 2026
Roughly 200 to 230 securities class actions a year (215 in 2023; 225 in 2024)	Cornerstone Research & Stanford Securities Class Action Clearinghouse, Securities Class Action Filings: 2024 Year in Review	Checked, Oct 2026 (range widened)

Figure	Source	Status
About 1,340 National Priorities List sites (1,343 final, March 2026); about 950 Love Canal families	EPA, Current NPL Updates; EPA Region 2 on Love Canal	Checked, Oct 2026 (NPL count updated)
About 1 compensated per 1 million doses	HRSA, National Vaccine Injury Compensation Program Data & Statistics (Oct. 2026)	Checked, Oct 2026
More than 5 million workers since 1974; WARN thresholds	Department of Labor, TAA program inventory; 29 U.S.C. § 2101(a)(1) (coverage), § 2102(a) (60 days' notice)	Checked, Oct 2026 (year and pinpoint corrected)
$500 per violation; up to $1,500 at the court's discretion for wilful violations	47 U.S.C. § 227(b)(3)(B)–(C)	Checked, Oct 2026
More than 100 deaths, 107 by the standard count (Elixir Sulfanilamide); thousands of children, about 10,000 by the standard estimate (thalidomide)	FDA history ("more than 100"; "thousands"); Wax, Annals of Internal Medicine (1995) for 107; the 10,000 figure is from secondary literature, not FDA	Checked, Oct 2026 (attribution corrected)
3,400 deaths, 43,000 injuries, $7 billion (projected); $71 million and about 140,000 to 144,000 people (Three Mile Island)	AEC, WASH-740 (1957); NRC Backgrounder, Nuclear Insurance and Disaster Relief ($71 million); Kemeny Commission and Dickinson College TMI archive (evacuation)	Checked, Oct 2026 (evacuation source corrected)
C$6.04 million claimed, C$3 million paid	Canada, Statement of Claim (Jan. 23, 1979) ¶¶ 8, 24; Canada–USSR Protocol (Apr. 2, 1981), art. I	Checked, Oct 2026
60,000 postings a day; at least 2 million subscribers	Stratton Oakmont, Inc. v. Prodigy Services Co., 1995 WL 323710, at *1, *3 (N.Y. Sup. Ct. 1995)	Checked, Oct 2026
2,977 killed	National September 11 Memorial & Museum / NYC Office of Chief Medical Examiner. The 9/11 Commission Report gives component figures only (more than 2,600 at the World Trade Center, 125 at the Pentagon, 256 on the planes)	Checked, Oct 2026 (source corrected)
About 7 million reported COVID-19 deaths	WHO COVID-19 Dashboard (cumulative reported deaths just over 7 million)	Checked, Oct 2026
$4.88 million average breach cost (global average)	IBM & Ponemon Institute, Cost of a Data Breach Report 2024	Checked, Oct 2026
US$5,000 sanction; C$812.02 award	Mata v. Avianca, Inc., 678 F. Supp. 3d 443 (S.D.N.Y. 2023); Moffatt v. Air Canada, 2024 BCCRT 149	Checked, Oct 2026
13% relative employment decline, ages 22 to 25 (July 2025 data)	Brynjolfsson, Chandar & Chen, "Canaries in the Coal Mine?" (Aug. 2025); the Aug. 2026 update reports 16% to 19%	Checked, Oct 2026 (dated)
0.15% of weekly users; about 800 million weekly users	OpenAI, "Strengthening ChatGPT's responses in sensitive conversations" (Oct. 27, 2025) for 0.15%; OpenAI DevDay keynote (Oct. 6, 2025) for 800 million	Checked, Oct 2026

Citation

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@online{margolin2026precedents,
  author = {Harrison C. Margolin},
  title  = {The Ratio Method report: Precedents},
  year   = {2026},
  note   = {Independent report; figures cite-checked October 2026}
}
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Footnotes

1. Harrison C. Margolin & Grant H. Frazier, *The Ratio Method: Addressing Complex Tort Liability in the Fourth Industrial Revolution*, 52 St. Mary's L.J. 679 (2021).

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